Snowflake and PowerBI Integration

Adding New Columns, New Measures and Making Reports Documentation

1. CREATE WAREHOUSE

```
CREATE WAREHOUSE SRINIVAS_SALES_WH
WAREHOUSE_SIZE = 'X-SMALL'
AUTO_SUSPEND = 60
AUTO_RESUME = TRUE
INITIALLY_SUSPENDED = TRUE;
```

2. CREATE DATABASE

```
CREATE DATABASE SRINIVAS_SALES_DB;
```

3. CREATE SCHEMA

```
CREATE SCHEMA SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA;
```

4. USE WAREHOUSE, DATABASE AND SCHEMA

```
USE WAREHOUSE SRINIVAS_SALES_WH;
USE DATABASE SRINIVAS_SALES_DB;
USE SCHEMA SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA;
```

5. CREATE TABLE

```
CREATE TABLE SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES
(
   ORDER_ID NUMBER,
   CUSTOMER_ID NUMBER,
   CUSTOMER_NAME VARCHAR(50),
   CITY VARCHAR(50),
   PRODUCT_CATEGORY VARCHAR(50),
   QUANTITY NUMBER,
   UNIT_PRICE NUMBER(10,2),
   ORDER_DATE DATE
);
```

6. INSERT 40 RECORDS

```
INSERT INTO SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES
(ORDER_ID, CUSTOMER_ID, CUSTOMER_NAME, CITY, PRODUCT_CATEGORY, QUANTITY, UNIT_PRICE,
ORDER_DATE)
VALUES
(1, 101, 'Rahul', 'Hyderabad', 'Electronics', 2, 25000, '2026-01-05'),
(2, 102, 'Priya', 'Bangalore', 'Clothing', 3, 1500, '2026-01-08'),
```

(3, 103, 'Arjun', 'Chennai', 'Furniture', 1, 15000, '2026-01-12'),

(4, 104, 'Sneha', 'Tirupati', 'Grocery', 5, 500, '2026-01-15'),

(5, 105, 'Kiran', 'Hyderabad', 'Clothing', 2, 2000, '2026-01-18'),

(6, 106, 'Ananya', 'Bangalore', 'Electronics', 1, 35000, '2026-01-20'),

(7, 107, 'Vijay', 'Chennai', 'Grocery', 4, 750, '2026-01-22'),

(8, 108, 'Meena', 'Tirupati', 'Furniture', 2, 12000, '2026-01-25'),

(9, 109, 'Rohit', 'Hyderabad', 'Electronics', 1, 45000, '2026-02-02'),

(10, 110, 'Divya', 'Bangalore', 'Clothing', 4, 1200, '2026-02-04'),

(11, 111, 'Suresh', 'Chennai', 'Furniture', 2, 18000, '2026-02-06'),

(12, 112, 'Pooja', 'Tirupati', 'Grocery', 6, 400, '2026-02-08'),

(13, 113, 'Manoj', 'Hyderabad', 'Clothing', 3, 1800, '2026-02-10'),

(14, 114, 'Keerthi', 'Bangalore', 'Electronics', 2, 22000, '2026-02-12'),

(15, 115, 'Naveen', 'Chennai', 'Grocery', 5, 600, '2026-02-14'),

(16, 116, 'Swathi', 'Tirupati', 'Furniture', 1, 25000, '2026-02-16'),

(17, 117, 'Ajay', 'Hyderabad', 'Electronics', 3, 18000, '2026-02-18'),

(18, 118, 'Lakshmi', 'Bangalore', 'Clothing', 2, 2500, '2026-02-20'),

(19, 119, 'Varun', 'Chennai', 'Furniture', 1, 30000, '2026-02-22'),

(20, 120, 'Deepa', 'Tirupati', 'Grocery', 8, 350, '2026-02-25'),

(21, 121, 'Ramesh', 'Hyderabad', 'Clothing', 5, 1400, '2026-03-02'),

(22, 122, 'Harika', 'Bangalore', 'Electronics', 1, 55000, '2026-03-04'),

(23, 123, 'Praveen', 'Chennai', 'Grocery', 3, 900, '2026-03-06'),

(24, 124, 'Sanjana', 'Tirupati', 'Furniture', 2, 16000, '2026-03-08'),

(25, 125, 'Gopal', 'Hyderabad', 'Electronics', 2, 30000, '2026-03-10'),

(26, 126, 'Neha', 'Bangalore', 'Clothing', 3, 2200, '2026-03-12'),

(27, 127, 'Tarun', 'Chennai', 'Furniture', 1, 22000, '2026-03-14'),

(28, 128, 'Bhavya', 'Tirupati', 'Grocery', 7, 450, '2026-03-16'),

(29, 129, 'Surya', 'Hyderabad', 'Electronics', 1, 65000, '2026-03-18'),

(30, 130, 'Madhavi', 'Bangalore', 'Clothing', 4, 1600, '2026-03-20'),

(31, 131, 'Chandu', 'Chennai', 'Grocery', 10, 300, '2026-04-02'),

(32, 132, 'Teja', 'Tirupati', 'Furniture', 3, 14000, '2026-04-04'),

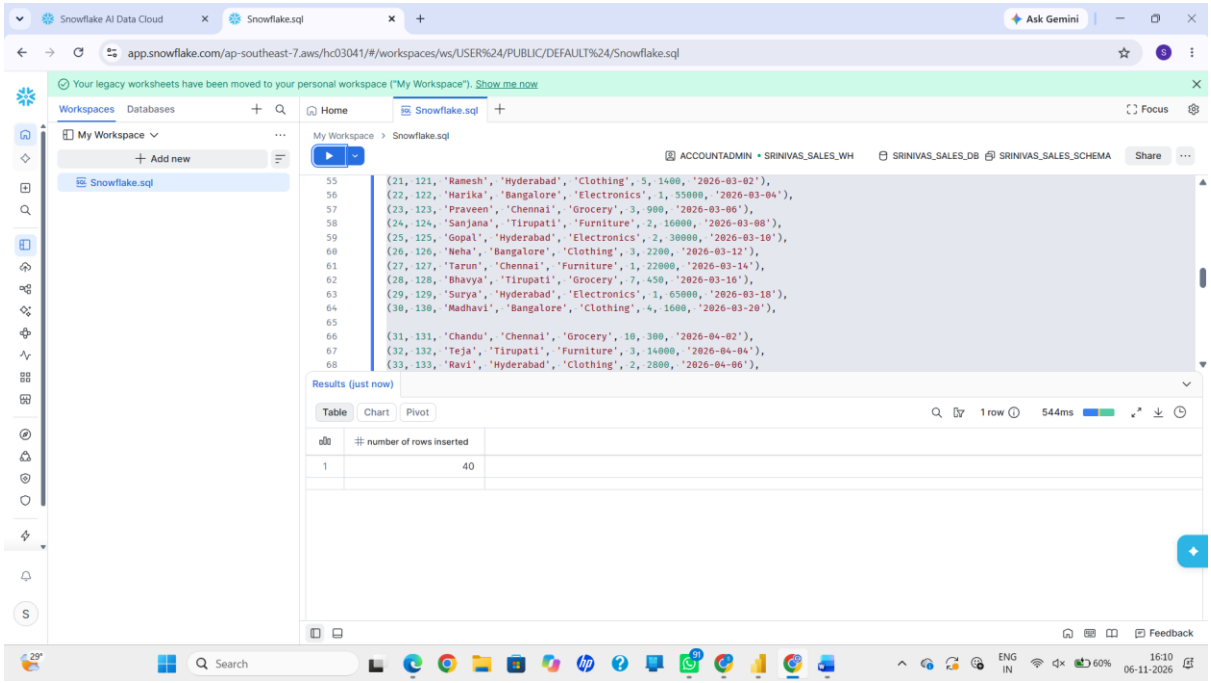
(33, 133, 'Ravi', 'Hyderabad', 'Clothing', 2, 2800, '2026-04-06'),

(34, 134, 'Sowmya', 'Bangalore', 'Electronics', 2, 40000, '2026-04-08'),

(35, 135, 'Karthik', 'Chennai', 'Furniture', 2, 20000, '2026-04-10'),

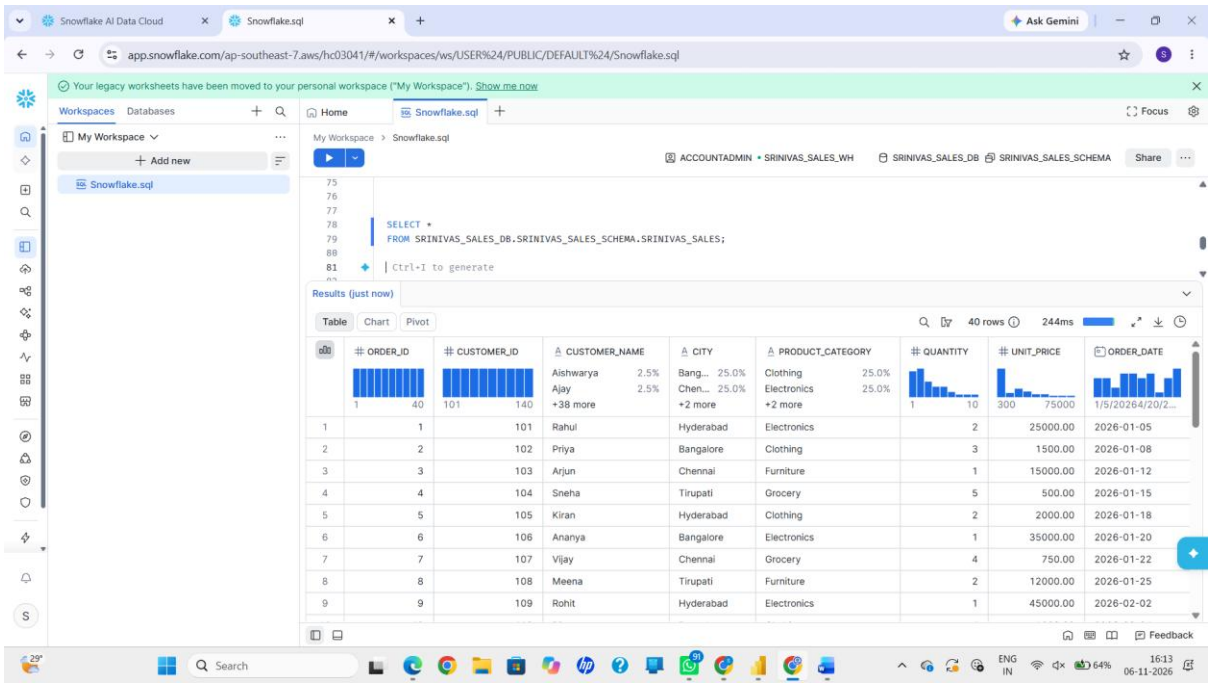
(36, 136, 'Aishwarya', 'Tirupati', 'Grocery', 5, 650, '2026-04-12'),

```
(37, 137, 'Mohan', 'Hyderabad', 'Electronics', 1, 75000, '2026-04-14'),
(38, 138, 'Sravani', 'Bangalore', 'Clothing', 6, 1300, '2026-04-16'),
(39, 139, 'Pavan', 'Chennai', 'Grocery', 4, 800, '2026-04-18'),
(40, 140, 'Nithya', 'Tirupati', 'Furniture', 1, 35000, '2026-04-20');
```



7. VIEW ALL RECORDS

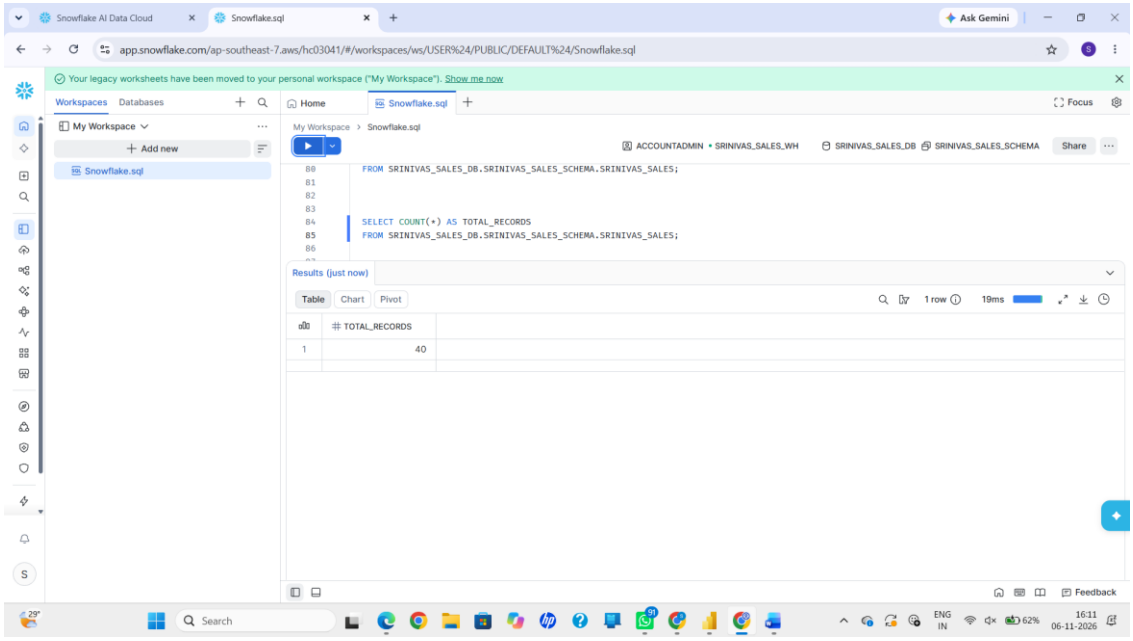
```
SELECT * FROM SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES;
```



8. VERIFY TOTAL RECORD COUNT

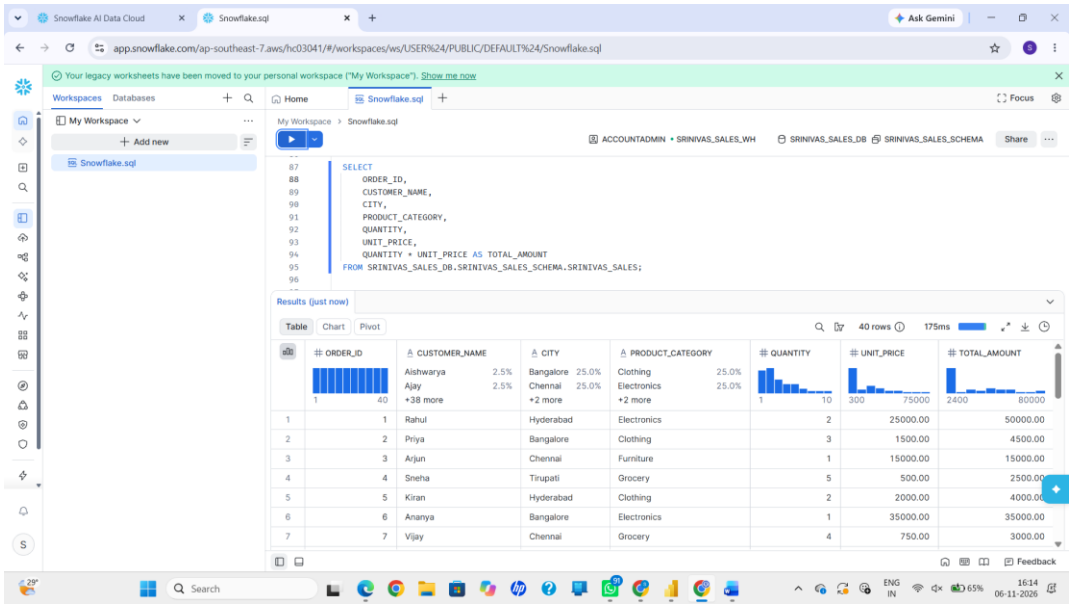
```
SELECT COUNT(*) AS TOTAL_RECORDS
```

```
FROM SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES;
```



9. CALCULATE TOTAL SALES

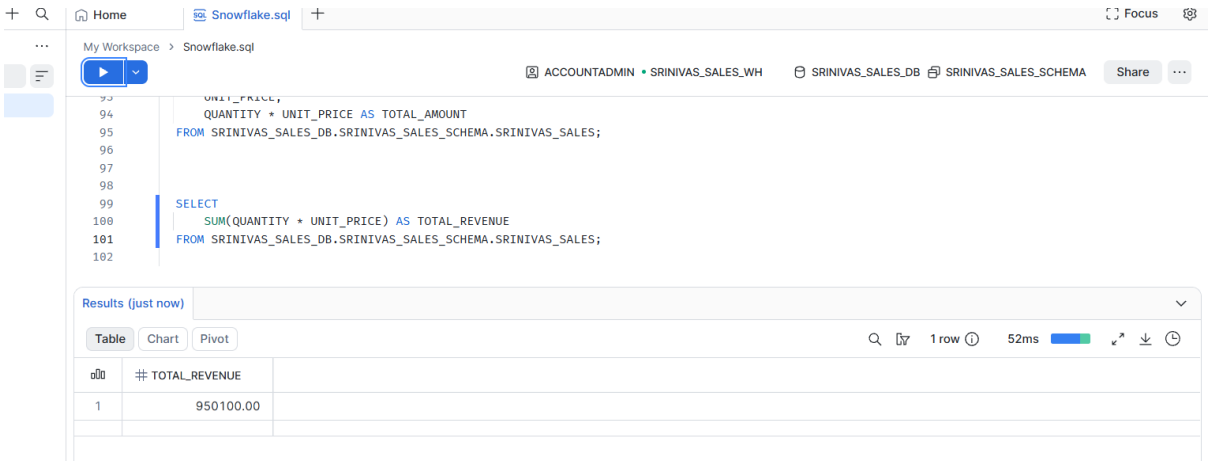
```
SELECT
    SALE_ID,
    CUSTOMER_NAME,
    CITY,
    PRODUCT_CATEGORY,
    QUANTITY,
    UNIT_PRICE,
    QUANTITY * UNIT_PRICE AS TOTAL_AMOUNT
FROM SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES;
```



10. TOTAL REVENUE

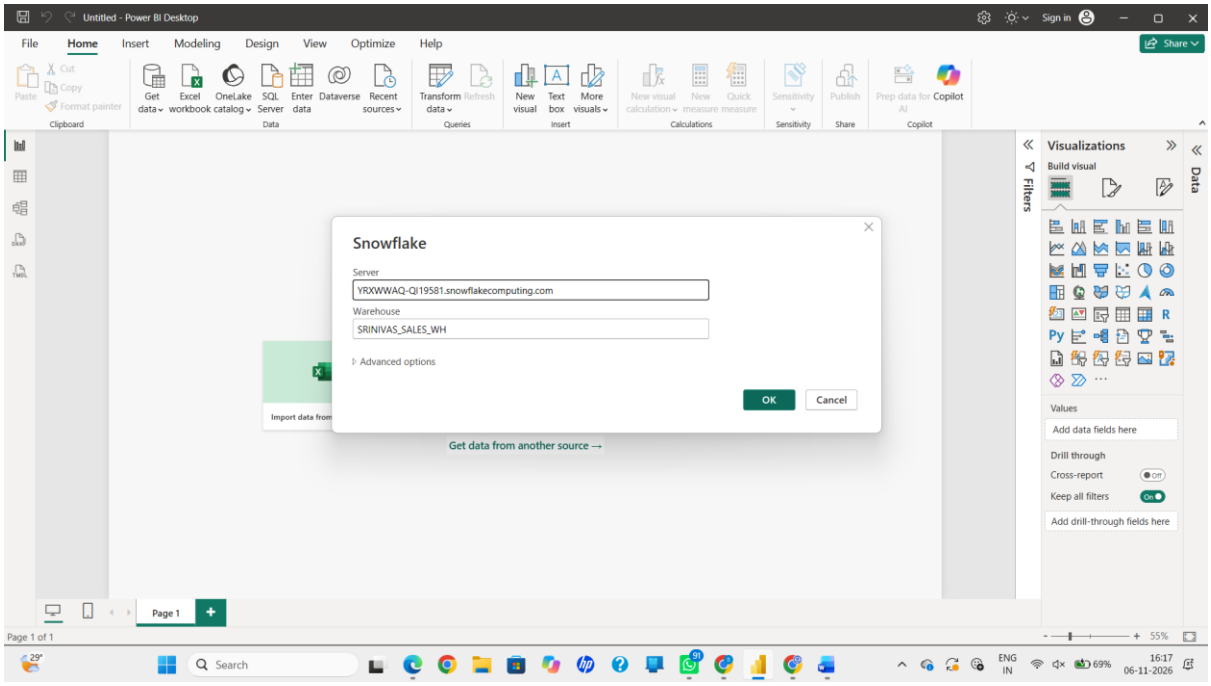
```
SELECT SUM(QUANTITY * UNIT_PRICE) AS TOTAL_REVENUE

FROM SRINIVAS_SALES_DB.SRINIVAS_SALES_SCHEMA.SRINIVAS_SALES;
```

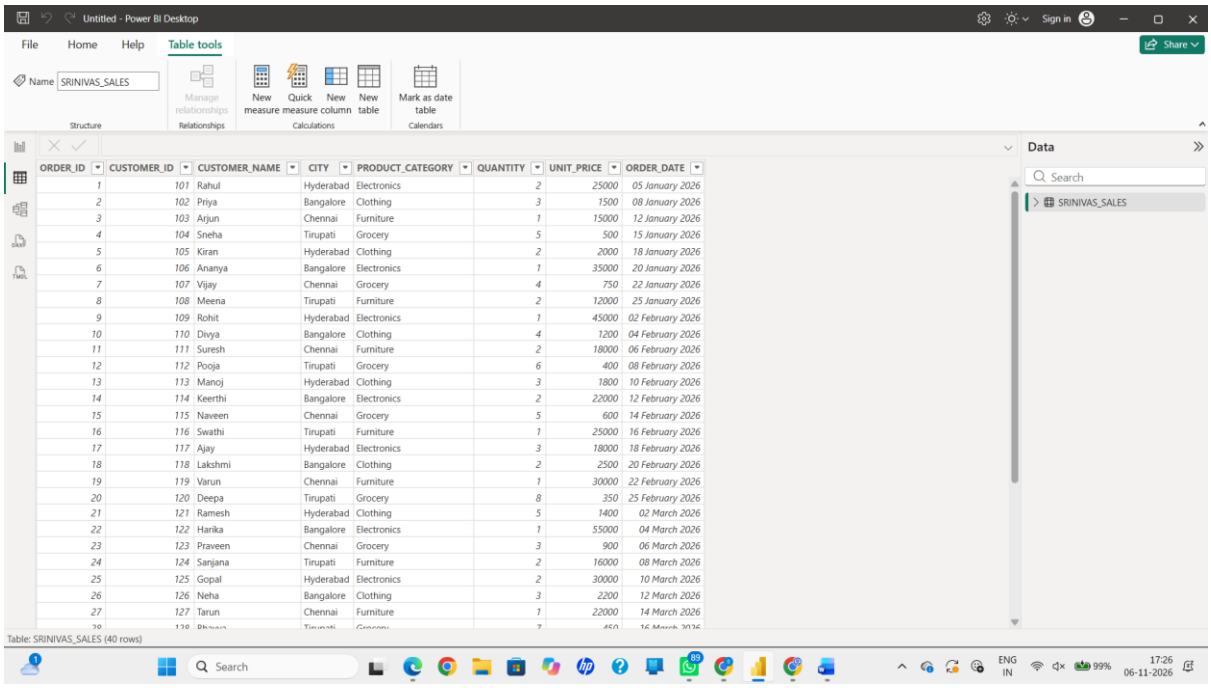


PowerBI Platform

Connecting Snowflake from PowerBI



Before Adding New Columns:



ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME	CITY	PRODUCT_CATEGORY	QUANTITY	UNIT_PRICE	ORDER_DATE
1	101	Rahul	Hyderabad	Electronics	2	25000	05 January 2026
2	102	Priya	Bangalore	Clothing	3	1500	08 January 2026
3	103	Arjun	Chennai	Furniture	1	15000	12 January 2026
4	104	Sneha	Tirupati	Grocery	5	500	15 January 2026
5	105	Kiran	Hyderabad	Clothing	2	2000	18 January 2026
6	106	Ananya	Bangalore	Electronics	1	35000	20 January 2026
7	107	Vijay	Chennai	Grocery	4	750	22 January 2026
8	108	Meena	Tirupati	Furniture	2	12000	25 January 2026
9	109	Rohit	Hyderabad	Electronics	1	45000	02 February 2026
10	110	Divya	Bangalore	Clothing	4	1200	04 February 2026
11	111	Suresh	Chennai	Furniture	2	18000	06 February 2026
12	112	Pooja	Tirupati	Grocery	6	400	08 February 2026
13	113	Manoj	Hyderabad	Clothing	3	1800	10 February 2026
14	114	Keerthi	Bangalore	Electronics	2	22000	12 February 2026
15	115	Naveen	Chennai	Grocery	5	600	14 February 2026
16	116	Swathi	Tirupati	Furniture	1	25000	16 February 2026
17	117	Ajay	Hyderabad	Electronics	3	18000	18 February 2026
18	118	Lakshmi	Bangalore	Clothing	2	2500	20 February 2026
19	119	Varun	Chennai	Furniture	1	30000	22 February 2026
20	120	Deepa	Tirupati	Grocery	8	350	25 February 2026
21	121	Ramesh	Hyderabad	Clothing	5	1400	02 March 2026
22	122	Harika	Bangalore	Electronics	1	55000	04 March 2026
23	123	Praveen	Chennai	Grocery	3	900	06 March 2026
24	124	Sanjana	Tirupati	Furniture	2	16000	08 March 2026
25	125	Gopal	Hyderabad	Electronics	2	30000	10 March 2026
26	126	Neha	Bangalore	Clothing	3	2200	12 March 2026
27	127	Tarun	Chennai	Furniture	1	22000	14 March 2026

<u>Adding New Columns:</u>

1. Sales Amount

Column name: Sales Amount

Objective: To calculate the total sales value generated from each order.

```
Sales Amount =

SRINIVAS_SALES[QUANTITY] * SRINIVAS_SALES[UNIT_PRICE]
```

2. Total Cost

Column name: Total Cost

Objective: To calculate the estimated total cost of products sold.

For Example, we assume that the product cost is 70% of its selling price.

```
Total Cost =

SRINIVAS_SALES[Sales Amount] * 0.70
```

3. Gross Profit

Column name: Gross Profit

Objective: To calculate the profit earned after deducting the product cost from the sales amount.

```
Gross Profit =

SRINIVAS_SALES[Sales Amount] - SRINIVAS_SALES[Total Cost]
```

4. Profit Category

Column name: Profit Category

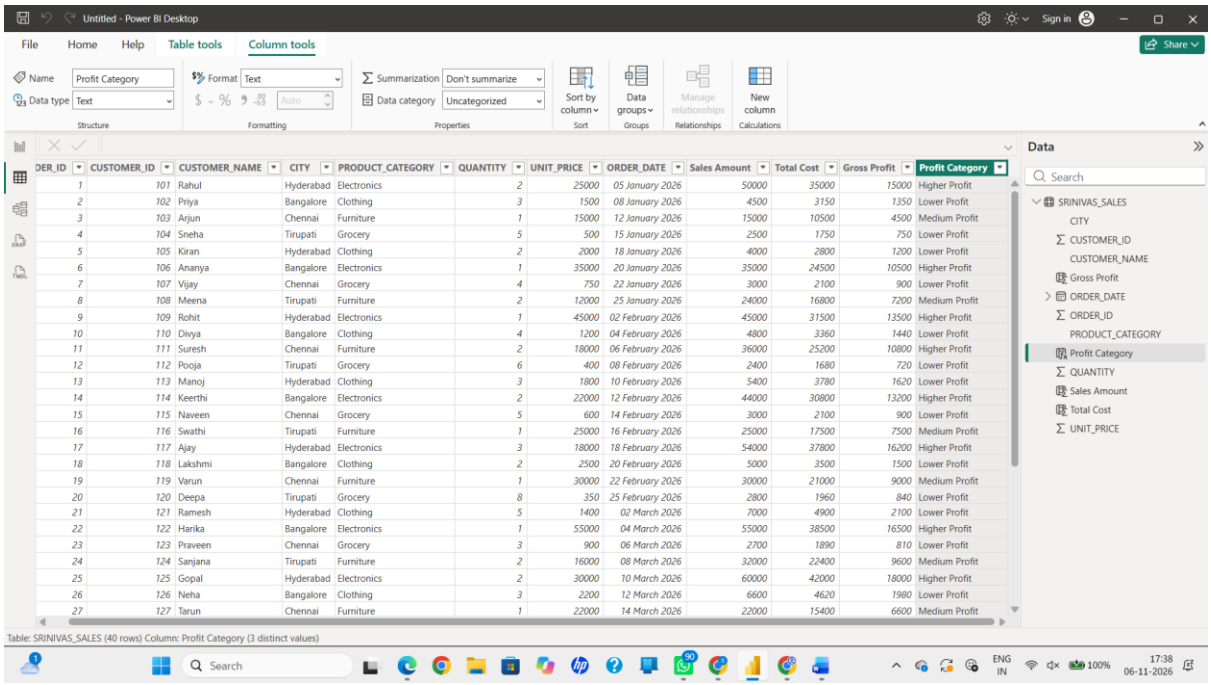
Objective: To classify transactions based on their gross profit.

We will use three categories:

- Higher Profit: Gross profit greater than or equal to ₹10,000.
- Medium Profit: Gross profit between ₹3,000 and ₹9,999.99.
- Lower Profit: Gross profit below ₹3,000.

```
Profit Category =
SWITCH(
   TRUE(),
   SRINIVAS_SALES[Gross Profit] >= 10000, "Higher Profit",
   SRINIVAS_SALES[Gross Profit] >= 3000, "Medium Profit",
   "Lower Profit"
)
```

After Adding 4 New Columns The Table View:



ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME	CITY	PRODUCT_CATEGORY	QUANTITY	UNIT_PRICE	ORDER_DATE	Sales Amount	Total Cost	Gross Profit	Profit Category
1	101	Rahul	Hyderabad	Electronics	2	25000	05 January 2026	50000	35000	15000	Higher Profit
2	102	Priya	Bangalore	Clothing	3	1500	08 January 2026	4500	3150	1350	Lower Profit
3	103	Arjun	Chennai	Furniture	1	15000	12 January 2026	15000	10500	4500	Medium Profit
4	104	Sneha	Tirupati	Grocery	5	500	15 January 2026	2500	1750	750	Lower Profit
5	105	Kiran	Hyderabad	Clothing	2	2000	18 January 2026	4000	2800	1200	Lower Profit
6	106	Ananya	Bangalore	Electronics	1	35000	20 January 2026	35000	24500	10500	Higher Profit
7	107	Vijay	Chennai	Grocery	4	750	22 January 2026	3000	2100	900	Lower Profit
8	108	Meena	Tirupati	Furniture	2	12000	25 January 2026	24000	16800	7200	Medium Profit
9	109	Rohit	Hyderabad	Electronics	1	45000	02 February 2026	45000	31500	13500	Higher Profit
10	110	Divya	Bangalore	Clothing	4	1200	04 February 2026	4800	3360	1440	Lower Profit
11	111	Suresh	Chennai	Furniture	2	18000	06 February 2026	36000	25200	10800	Higher Profit
12	112	Pooja	Tirupati	Grocery	6	400	08 February 2026	2400	1680	720	Lower Profit
13	113	Manoj	Hyderabad	Clothing	3	1800	10 February 2026	5400	3780	1620	Lower Profit
14	114	Keerthi	Bangalore	Electronics	2	22000	12 February 2026	44000	30800	13200	Higher Profit
15	115	Naveen	Chennai	Grocery	5	600	14 February 2026	3000	2100	900	Lower Profit
16	116	Swathi	Tirupati	Furniture	1	25000	16 February 2026	25000	17500	7500	Medium Profit
17	117	Ajay	Hyderabad	Electronics	3	18000	18 February 2026	54000	37800	16200	Higher Profit
18	118	Lakshmi	Bangalore	Clothing	2	2500	20 February 2026	5000	3500	1500	Lower Profit
19	119	Varun	Chennai	Furniture	1	30000	22 February 2026	30000	21000	9000	Medium Profit
20	120	Deepa	Tirupati	Grocery	8	350	25 February 2026	2800	1960	840	Lower Profit
21	121	Ramesh	Hyderabad	Clothing	5	1400	02 March 2026	7000	4900	2100	Lower Profit
22	122	Harika	Bangalore	Electronics	1	55000	04 March 2026	55000	38500	16500	Higher Profit
23	123	Praveen	Chennai	Grocery	3	900	06 March 2026	2700	1890	810	Lower Profit
24	124	Sanjana	Tirupati	Furniture	2	16000	08 March 2026	32000	22400	9600	Medium Profit
25	125	Gopal	Hyderabad	Electronics	2	30000	10 March 2026	60000	42000	18000	Higher Profit
26	126	Neha	Bangalore	Clothing	3	2200	12 March 2026	6600	4620	1980	Lower Profit
27	127	Tarun	Chennai	Furniture	1	22000	14 March 2026	22000	15400	6600	Medium Profit

Creating New Measures :

1. Total Sales

Objective: Calculate the total revenue generated from all sales transactions.

```
Total Sales =
```

```
SUMX(
    SRINIVAS_SALES,
    SRINIVAS_SALES[QUANTITY] *
    SRINIVAS_SALES[UNIT_PRICE]
)
```

Explanation : This measure calculates the overall sales revenue by multiplying the quantity sold by the unit price for every transaction and adding the results. It provides an overview of the total business revenue.

2. Total Customers

Objective: Count the number of unique customers.

```
Total Customers =
DISTINCTCOUNT(SRINIVAS_SALES[CUSTOMER_ID])
```

Explanation : This measure identifies the total number of unique customers who have purchased products. It helps us understand the customer base and monitor customer reach.

3. Total Orders

Objective: Count the total number of orders.

```
Total Orders =
DISTINCTCOUNT(SRINIVAS_SALES[ORDER_ID])
```

Explanation : This measure calculates the total number of unique sales orders recorded in the dataset. It helps track transaction volume and understand business activity.

4. Average Order Value (AOV)

Objective: Calculate the average revenue generated per order.

```
Average Order Value =
DIVIDE(
    [Total Sales],
    [Total Orders],
    0
)
```

Explanation : This measure calculates the average amount spent on each order by dividing total sales by the total number of orders. It helps understand the average customer spending per transaction.

5. Total Gross Profit

Objective: Calculate the overall gross profit generated from all sales transactions.

```
Total Gross Profit =
SUM(SRINIVAS_SALES[Gross Profit])
```

Explanation : This measure calculates the total gross profit by adding the gross profit earned from individual sales transactions. It helps understand the overall profitability of the business and compare profit performance across different cities and product categories.

6. Total Margin

Objective: Calculate the total profit margin percentage.

```
Total Margin =
DIVIDE(
    [Total Gross Profit],
    [Total Sales],
    0
)
```

Explanation : This measure calculates the percentage of sales revenue remaining as gross profit after deducting product costs. It helps evaluate the profitability of the business.

7. Profit Margin

For this dashboard, we can use Profit Margin as a percentage measure at the transaction level, aggregated correctly across the selected data.

```
Profit Margin =
DIVIDE(
    [Total Gross Profit],
    [Total Sales],
    0
)
```

Explanation : This measure shows the percentage of revenue retained as gross profit. A higher profit margin indicates that the business retains more profit from its sales.

8. Sales Target

Objective: Define the target revenue that the business wants to achieve.

```
Sales Target =
100000
```

Explanation : This measure defines the sales target as ₹1,00,000. It acts as a benchmark for comparing actual sales performance against the expected business goal.

9. Target Achievement

Objective: Calculate how much of the sales target has been achieved.

```
Target Achievement =
DIVIDE(
    [Total Sales],
    [Sales Target],
    0
)
```

Explanation : This measure calculates the percentage of the sales target achieved by comparing actual sales with the predefined target. It helps us understand whether the business is progressing towards its expected revenue goal.

For example, if actual sales are ₹75,000 against a target of ₹1,00,000, the achievement is 75%.

10. Target Not Achieved %

Objective: Calculate the percentage of the target that is still pending.

```
Target Not Achieved % =
MAX(
    0,
    DIVIDE(
        [Sales Target] - [Total Sales],
        [Sales Target],
        0
    )
)
```

Explanation : This measure calculates the percentage of the sales target that has not yet been achieved. It helps identify the remaining gap between actual sales and the expected target.

For example, if sales reach ₹75,000 against a target of ₹1,00,000, the remaining target gap is 25%.

11. Target Shortfall

Objective: Calculate the actual amount of revenue remaining to reach the target.

```
Target Shortfall =
MAX(
    0,
    [Sales Target] - [Total Sales]
)
```

Explanation : This measure calculates the difference between the sales target and actual sales. It indicates how much additional revenue the business needs to generate to achieve its target.

12. Target Status

Objective: Display whether the business has achieved its sales target.

```
Target Status =
IF(
    [Total Sales] >= [Sales Target],
    "Target Achieved",
    "Not Achieved"
)
```

Explanation : This measure checks actual sales against the predefined sales target and displays the current performance status. It provides a simple indication of whether the business has successfully reached its revenue goal.

Visualization Charts

1. Sales by City – Clustered Bar Chart

Objective: To identify which cities generate the highest sales revenue and compare the sales performance across different locations.

Explanation: This chart displays the total sales revenue generated by each city. It helps us compare the performance of Hyderabad, Bangalore, Chennai, and Tirupati. By analyzing the chart, we can identify the cities contributing the most revenue and the locations where sales performance needs improvement.

2. Sales by Product – Clustered Column Chart

Objective: To compare the sales revenue generated by different product categories and identify the products contributing the most to overall sales.

Explanation: This chart represents the total revenue generated from different product categories, such as Electronics, Clothing, Furniture, and Grocery. It helps us understand which product categories perform better and which categories contribute less to total revenue.

3. Monthly Sales Trend – Line Chart

Objective: To analyze how sales revenue changes over time and identify sales growth, declining periods, and overall business trends.

Explanation: This chart displays the monthly sales revenue using order dates. It helps us understand how sales performance changes from month to month. By observing the trend, we can identify periods with higher and lower sales and understand the overall revenue pattern.

4. Sales by Category – Donut Chart

Objective: To understand the contribution of each product category to overall sales revenue and identify the categories with the highest sales share.

Explanation: This chart represents the percentage contribution of each product category to total sales revenue. It helps us understand how much each category contributes to the overall business performance. The visualization makes it easier to compare the sales distribution among Electronics, Clothing, Furniture, and Grocery.

After Hiding Unwanted Fields the Report View:

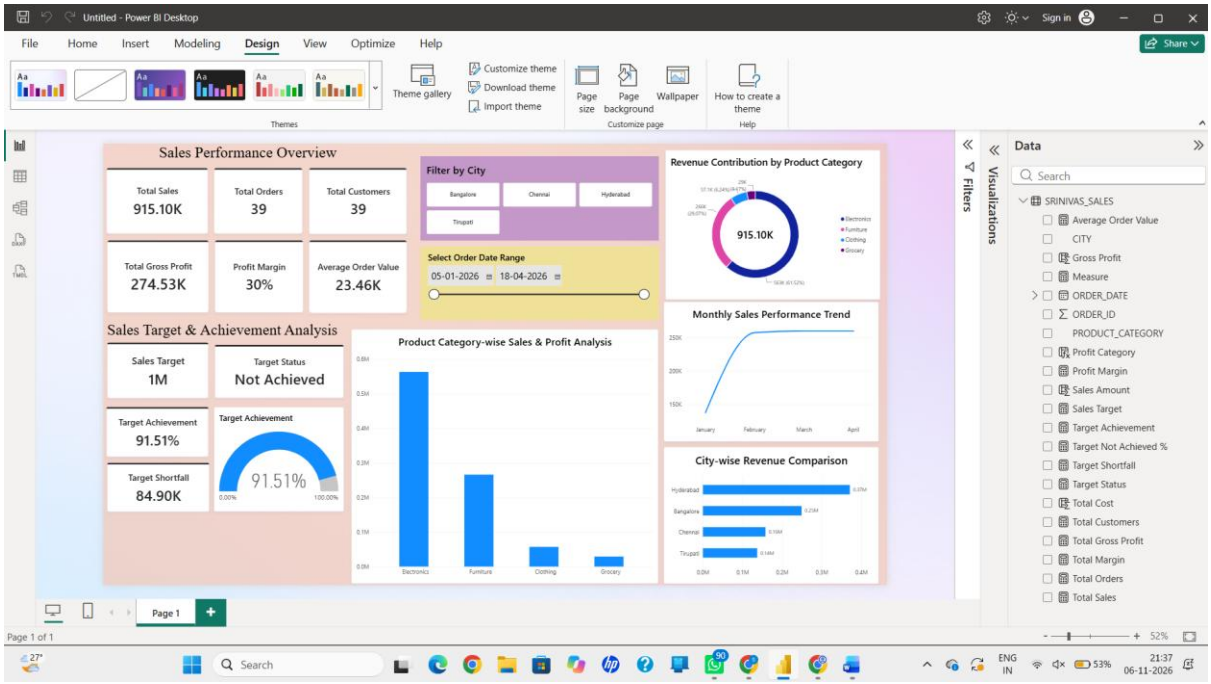
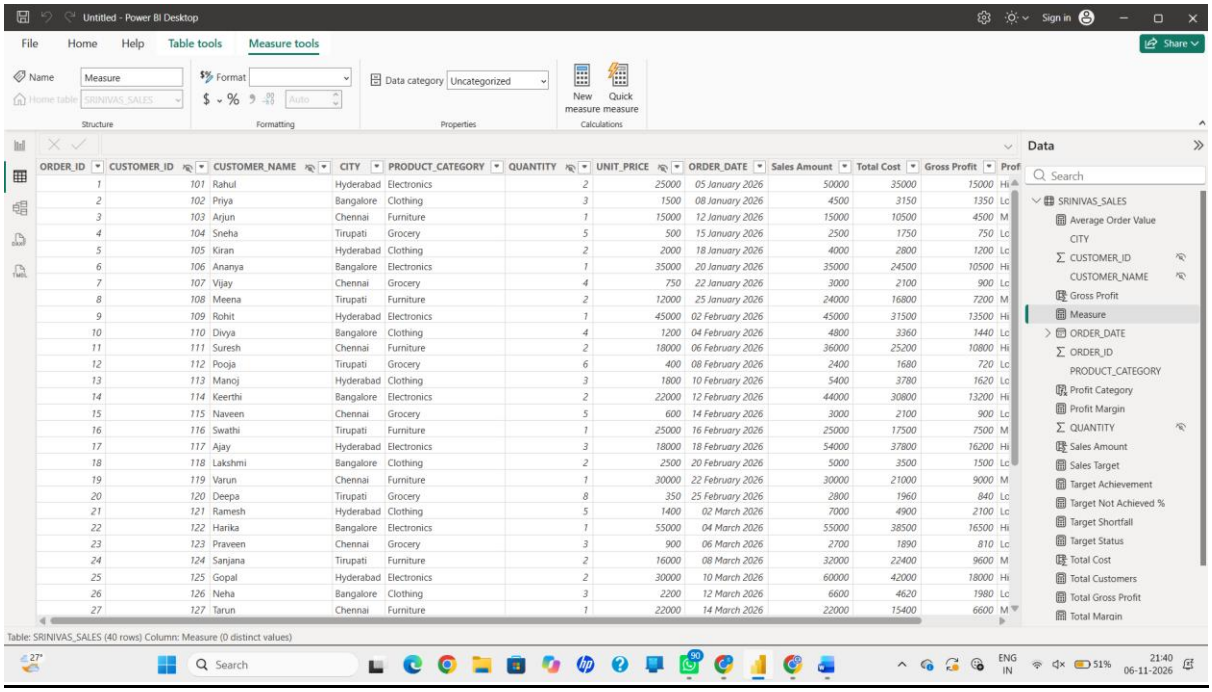


Table View Looks Same even if we hide some fields in Report View:



Final Report :

